

Machine Learning and the Physical World

Lecture 10 : Design of Experiments

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Model-free RL

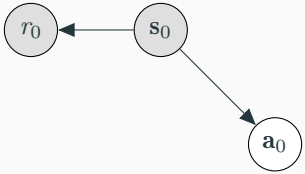
Reinforcement Learning



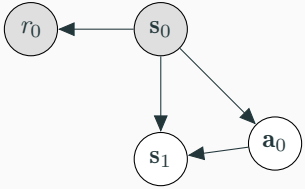
Reinforcement Learning



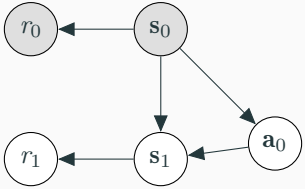
Reinforcement Learning



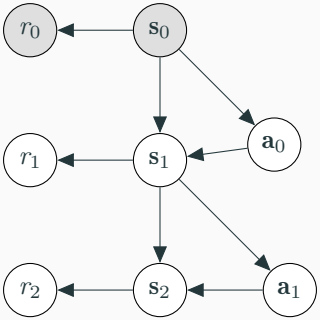
Reinforcement Learning



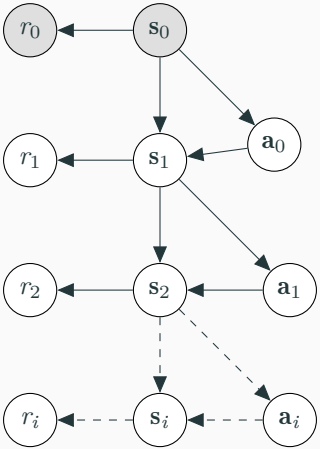
Reinforcement Learning



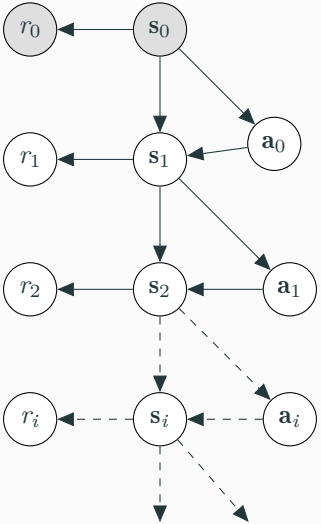
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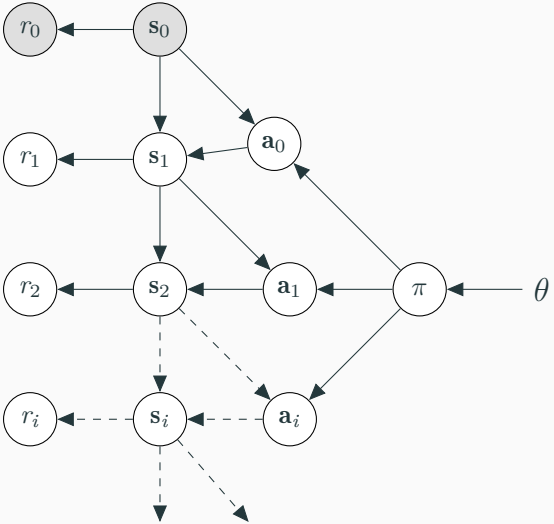
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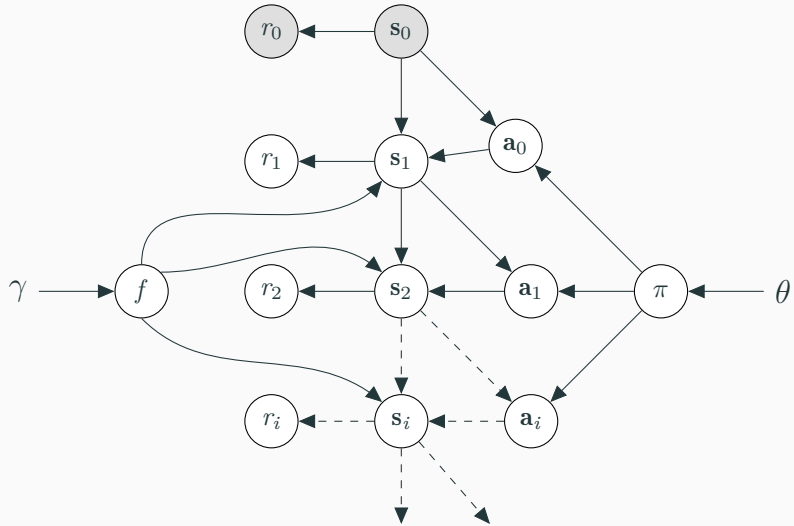
Reinforcement Learning



Reinforcement Learning



Reinforcement Learning



Model

$$p(\mathbf{s}, \mathbf{a}, \mathbf{r} \mid f, \pi)$$

Learning Task

$$p(f, \pi \mid \mathbf{s}, \mathbf{a}, \mathbf{r}) = \frac{p(\mathbf{s}, \mathbf{a}, \mathbf{r}, f, \pi)}{p(\mathbf{s}, \mathbf{a}, \mathbf{r})}$$

Marginal Likelihood

$$\int \prod_{t=1}^{\infty} p(r_t \mid s_t) p(s_t \mid f, a_t) p(a_t \mid s_{t-1}, \pi) p(f) p(\pi) \mathrm{d}f \mathrm{d}\pi$$

Model-free Reinforcement Learning

$$\mathbb{E}[r_t] = \int \int p(r_t \mid s_{t+1}) p(s_t + 1 \mid s_t, a_t) p(a_t \mid s_t) \mathrm{d}a_t \mathrm{d}s_{t+1}$$

$$\pi_* = \max_\pi \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \mid a_t \sim p(a_t \mid s_t)\right]$$

- We want to avoid explicitly learning a policy and a reward model
- Directly maximise the expected reward

Q-Learning

- Define value function V,

$$V_\pi(s_0) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \mid a_t \sim \pi(s_t)\right]$$

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- Lets move out the first action and define the function,

$$\tilde{V}_\pi(s_0, a_0) = \mathbb{E}[r_0 \mid a_0] + \mathbb{E}\left[\sum_{t=1}^{\infty} \gamma^t r_t \mid a_t \sim \pi(s_t)\right]$$

Q-Learning

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- This we will refer to as the Q-function,

$$Q_\pi(s_0, a_0) = \tilde{V}_\pi(s_0, a_0)$$

Q-Learning

- How should we pick our policy?

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- The greedy policy is optimal (Bellman Optimality Equation)

$$\pi_*(s) = \text{argmax}_{a \in A}\, Q_*(s, a)$$

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- The optimal Q function,

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Lets re-arrange things

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Lets re-arrange things

$$
\begin{aligned}
Q_*(s, a) &= \mathbb{E}[r_0 \mid a_0] + \mathbb{E}\left[\sum_{t=1}^{\infty} \gamma^t r_t \mid a_t \sim \pi_*(s_t)\right] \\
&= \mathbb{E}[r_0 \mid a_0] + \gamma \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_{t+1} \mid a_{t+1} \sim \pi_*(s_{t+1})\right]
\end{aligned}
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&= \mathbb{E}[r_0 \mid a_0] + \gamma \cdot Q_*(s_1, \pi_*(s_1)) \\
&= \mathbb{E}[r_0 \mid a_0] + \gamma \operatorname{argmax}_{a \in A} Q(s_1, a)
\end{aligned}
$$

Learning the Q

$$\theta_* = \text{argmin}_\theta \left(\text{Q}_\theta(\text{s}, \text{a})\text{-Q}_*(\text{s}, \text{a})\right)^2$$

- Can we learn a θ such that we can approximate the optimal Q?
- For any $\{s_0, s_1, a_0, r\}$ we will receive an error

Learning the Q

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Learning the Q

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&= \operatorname{argmin}_\theta \left(\underbrace{\left(Q_\theta(s,a)\text{-}\gamma \operatorname{argmax}_{a_1 \in A} Q_\theta(s_1,a_1)\right)}_{\Delta \mathrm{Q}} \text{-}\mathbb{E}[\mathrm{r}_0 \mid \mathrm{a}_0] \right)^2
\end{aligned}
$$

- Can we learn a θ such that we can approximate the optimal Q?
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Challenge of RL

- Model based RL
  - allows us to input our knowledge into the system
  - allows for (more) principled acquisition strategies
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- Model based RL
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  - very challenging inference task
- Model free RL
  - simple to implement and relatively cheap
  - hard to device principle acquisition strategies
  - often very data inefficient

Summary

- Reinforcement learning is often explained as a thing in itself
- It is not, its just a very specific, recursive probabilistic model
- We apply RL to problems where it is infeasible to acquire all data at the same time
- Just as BO the data-acquisition therefore has to be part of the inference mechanism
- There are many things where RL is a suitable approach

Design of Experiments

Task 1 encode your knowledge in a model

Task 2 randomly sample some data

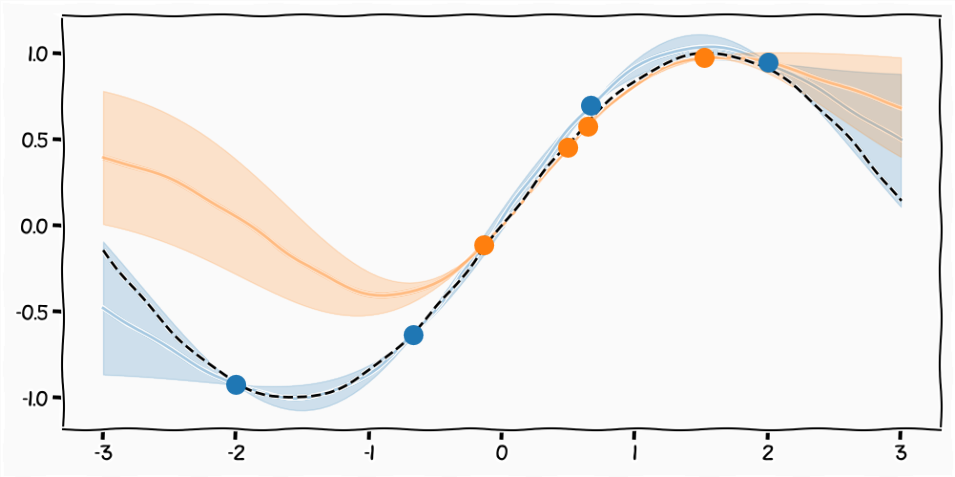
Task 3 update model with data and evaluate quantity of interest

Task 4 evaluate acquisition function to decide where to query data

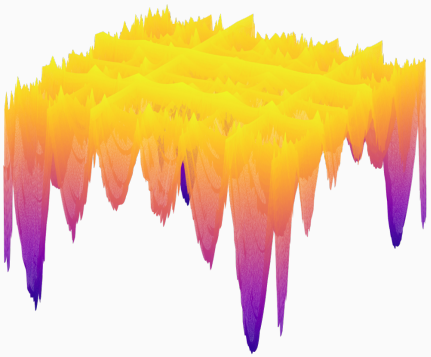
Task 5 update model with data

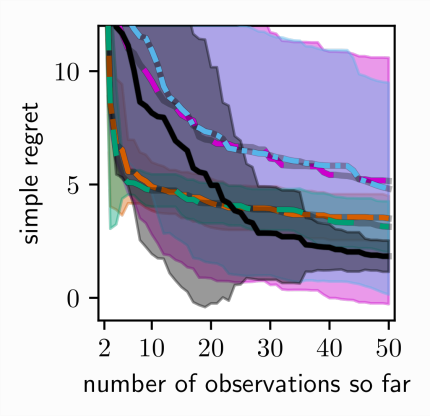
Task 6 loop 3-5 till budget is gone

Effect of Initial Design

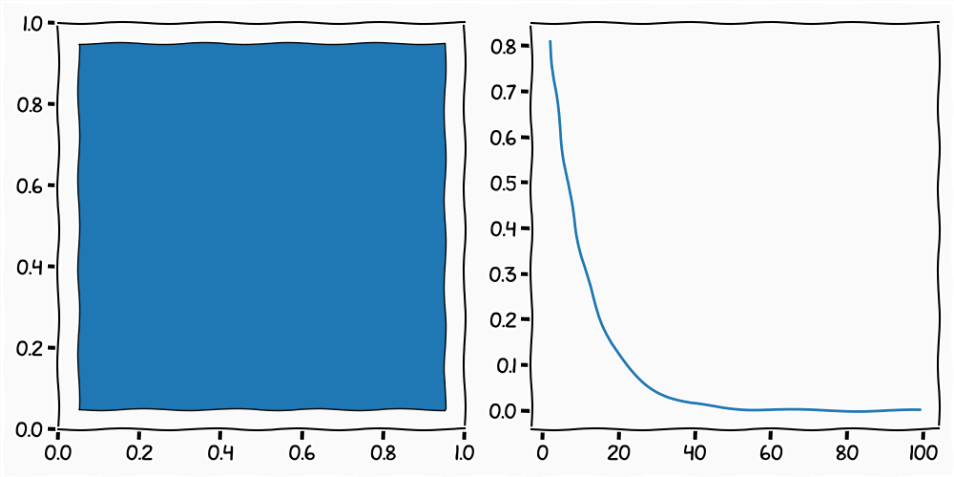


Effect of Initial Design

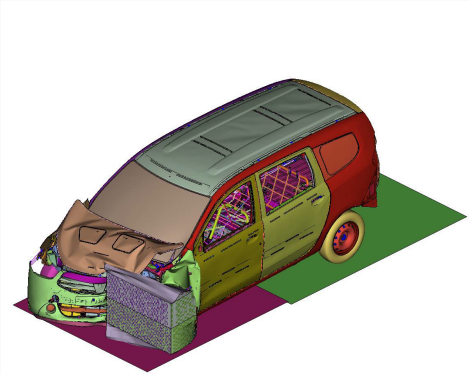




High-dimensional Spaces

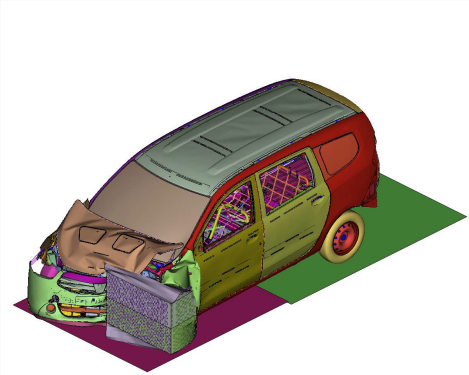


Cost of dimensionality



- 50 input parameters
- One FEM simulation takes 1 minute

Cost of dimensionality



- 50 input parameters
- One FEM simulation takes 1 minute
- Cost of min/max combinations

$$2^{50}\text{min} \approx 2 \times 10^{9}\text{years}$$

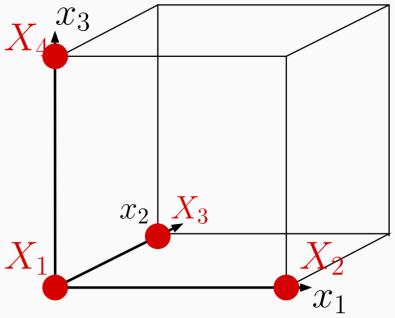
Design of Experiments

- How can we design a set of experiments that allows us to build a good model?
- Can we think of how to query data in order to be able to draw specific conclusions?

Model Free Design of Experiments

One-at-the-time Designs

x_1	x_2	x_3
0	0	0
1	0	0
0	1	0
0	0	1



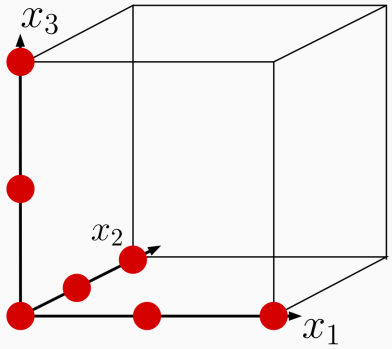
Linear Effects

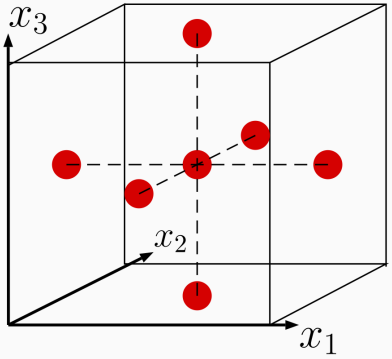
- requires only $d + 1$ observations
- easy to interpret
- can only show linear effects

$$f(x) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

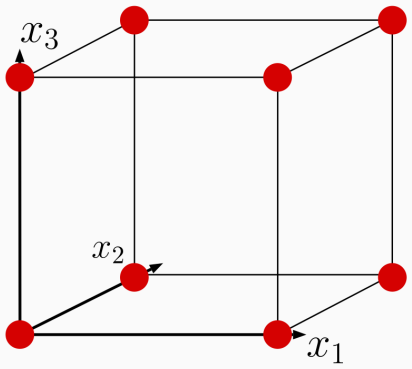
- does not cover space particularly well

Starshaped Designs





Factorial Designs

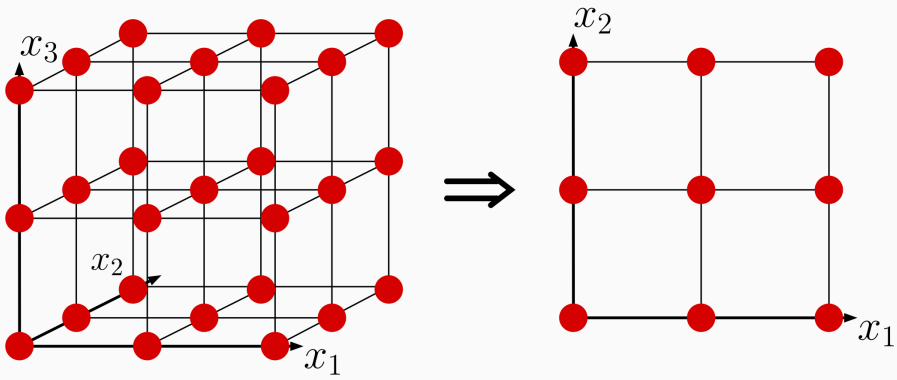


- Consider all combinations $x_i \in \{0, 1\}$
- Allows all interactions terms

$$\beta_0 + \sum_k \beta_k x_k + \sum_{j,k} \beta_{jk} x_j x_k + \beta_{123} x_1 x_2 x_3$$

- Number of evaluations grows very quickly

Projection of Designs



Classic Designs

- Pro
  - Very easy to use
  - Well suited for linear models
- Cons
  - Number of evaluations not flexible
  - Scales badly with dimensions
  - Projection issues

Optimal Designs for Models

$$y = \mathbf{X}\boldsymbol{\beta} + \epsilon$$
$$\epsilon \sim \mathcal{N}(0, \sigma^2\mathbf{I})$$

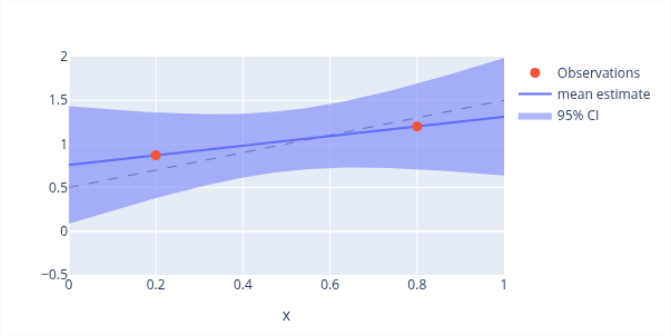
- We will refer to $\mathbf{X} \in \mathbb{R}^{n \times p}$ as a design matrix
- Least square estimator

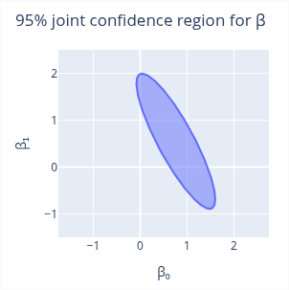
$$\hat{\beta} = \left(\mathbf{X}^{\mathrm{T}}\mathbf{X}\right)^{-1}\mathbf{X}^{\mathrm{T}}\mathbf{y}$$

- With distribution

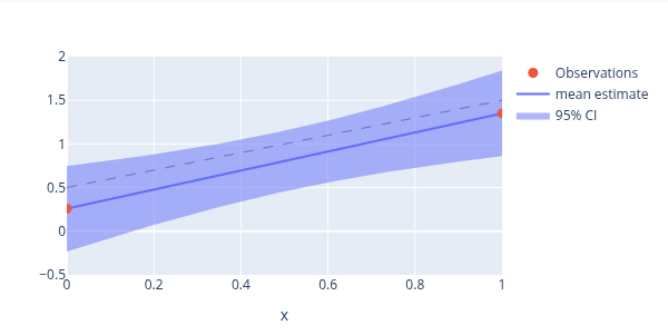
$$\hat{\beta} \sim \mathcal{N}\left(\beta, \sigma^2\left(\mathbf{X}^{\mathrm{T}}\mathbf{X}\right)^{-1}\right)$$

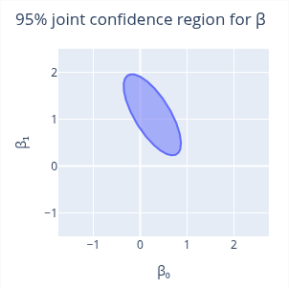
Linear Regression



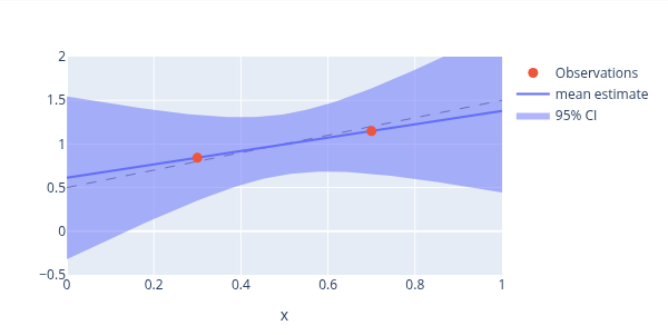


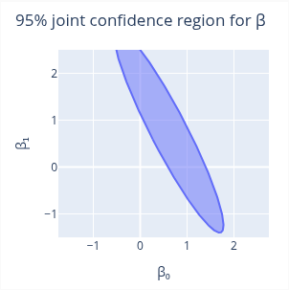
Linear Regression





Linear Regression





Design for Linear Regression

$$\mu(x) = B(x) \left(\mathbf{X}^{\mathrm{T}}\mathbf{X}\right)^{-1} \mathbf{X}^{\mathrm{T}}\mathbf{y}$$
$$\sigma(x) = \sigma^2 B(x) \left(\mathbf{X}^{\mathrm{T}}\mathbf{X}\right)^{-1} B(x)^{\mathrm{T}}$$
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- Prediction variance does not depend on the observations!

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- The distribution of the least square estimate does not depend on the observations!

Design for Linear Regression

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$$\hat{\beta} \sim \mathcal{N}\left(\beta, \sigma^2 \left(\mathbf{X}^{\mathrm{T}}\mathbf{X}\right)^{-1}\right)$$

- Prediction variance does not depend on the observations!
- The distribution of the least square estimate does not depend on the observations!
- *We can choose the experiment* $\mathbf{X}$ *to minimise either*

Criteria (estimate)

- D-optimality minimise the volume of the confidence in the estimate

$$\min_{\mathbf{X}} \det\left(\left(\mathbf{X}^{\mathrm{T}}\mathbf{X}\right)^{-1}\right) = \max_{\mathbf{X}} \det\left(\mathbf{X}^{\mathrm{T}}\mathbf{X}\right)$$

Criteria (estimate)

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- A-optimality minimise sum of variance of coefficients

$$\min_{\mathbf{X}} \operatorname{tr}\left(\mathbf{X}^{\mathrm{T}}\mathbf{X}\right)^{-1}$$

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- A-optimality minimise sum of variance of coefficients

$$\min_{\mathbf{X}} \operatorname{tr}\left(\mathbf{X}^{\mathrm{T}}\mathbf{X}\right)^{-1}$$

- E-optimality Minimise the largest eigenvalue

$$\min_{\mathbf{X}} \lambda_{\max}\left(\left(\mathbf{X}^{\mathrm{T}}\mathbf{X}\right)^{-1}\right)$$

Criteria (prediction)

- G-optimality minimise the maximum prediction variance

$$\min_{\mathbf{X}} \max_x \sigma^2 B(x) \left(\mathbf{X}^{\mathrm{T}}\mathbf{X}\right)^{-1} B(x)^{\mathrm{T}}$$

Criteria (prediction)

- G-optimality minimise the maximum prediction variance

$$\min_{\mathbf{X}} \max_x \sigma^2 B(x) \left(\mathbf{X}^{\mathrm{T}}\mathbf{X}\right)^{-1} B(x)^{\mathrm{T}}$$

- I-optimality minimise integrated variance

$$\min_{\mathbf{X}} \int B(x) \left(\mathbf{X}^{\mathrm{T}}\mathbf{X}\right)^{-1} B(x)^{\mathrm{T}} \mathrm{d}x$$

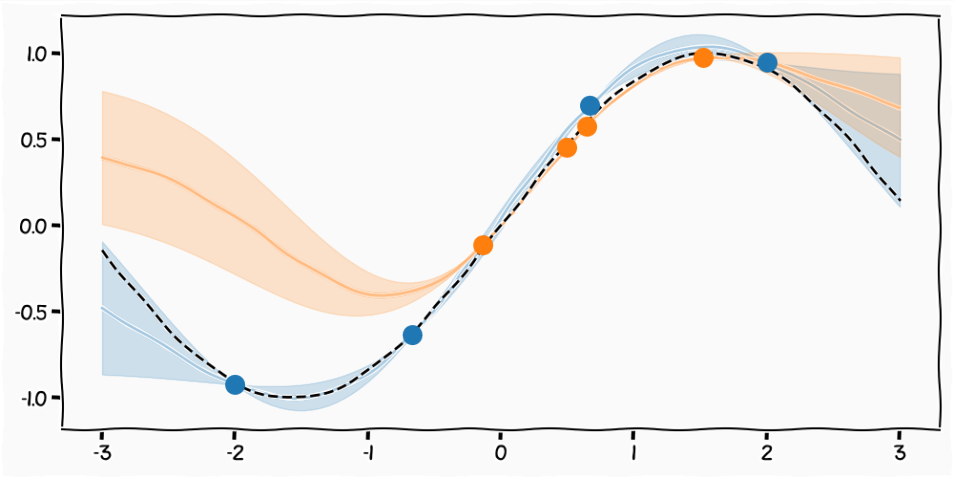
Equivalence Theorem (Keifer, Wolfowitz)

- The following three conditions are equivalent,
  - A design is D-optimal
  - A design is G-optimal
  - The maximum prediction variance is the same as the number of parameters p

Equivalence Theorem (Keifer, Wolfowitz)

- The following three conditions are equivalent,
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  - A design is G-optimal
  - The maximum prediction variance is the same as the number of parameters p
- Allows us to connect estimate and prediction optimality

Gaussian Processes



$$f(x) = \underbrace{k(x, \mathbf{X})k(\mathbf{X}, \mathbf{X})^{-1}f(\mathbf{X})}_{f_{\mathbf{X}}(x)} + \underbrace{f(x) - k(x, \mathbf{X})k(\mathbf{X}, \mathbf{X})^{-1}f(\mathbf{X})}_{f_{\mathbf{X}}^{\perp}(x)}$$

$$k(x, x') = \underbrace{k(x, \mathbf{X})k(\mathbf{X}, \mathbf{X})^{-1}k(\mathbf{X}, x')}_{k_{\mathbf{X}}(x,x')} + \underbrace{k(x, x') - k(x, \mathbf{X})k(\mathbf{X}, \mathbf{X})^{-1}k(\mathbf{X}, x')}_{k_{\mathbf{X}^{\perp}(x,x')}}$$

- Write the GP-prediction as a sum of two independent GPs

$$f(x) = \underbrace{k(x, \mathbf{X})k(\mathbf{X}, \mathbf{X})^{-1}f(\mathbf{X})}_{f_{\mathbf{X}}(x)} + \underbrace{f(x) - k(x, \mathbf{X})k(\mathbf{X}, \mathbf{X})^{-1}f(\mathbf{X})}_{f^{\perp}_{\mathbf{X}}(x)}$$

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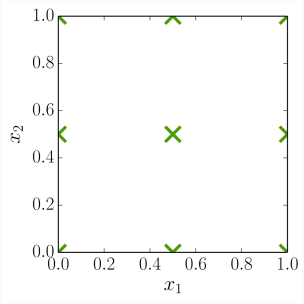
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- Estimate: D/A/E-optimality criteria on $k(\mathbf{X}, \mathbf{X})$

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- Write the GP-prediction as a sum of two independent GPs
- Estimate: D/A/E-optimality criteria on $k(\mathbf{X}, \mathbf{X})$
- Prediction: I/G-optimality applied to $k_{\mathbf{X}^{\perp}(x,x')}$

Known Covariance



- This design is neither I or G optimal
- There is no D-G equivalence for GP regression

$$\max_{\mathbf{X}} \det\left(k(\mathbf{X}, \mathbf{X})\right)$$

Unknown Covariance

Can we find a design that is good for estimating hyper-parameters?

- There are no strong theoretical results just ”rules of thumb”
- Variance estimates requires points to be far apart
- Length scale estimates requires points to be close together
- Good practice is to use a design with a variety of inter-point distances

Summary

- For linear models we have strong results
- Applicable to Generalise Linear Models which is a very strong model class
- For non-parametrics it is a different story
  - criteria is hard to compute
  - optimisation is not easy
  - we don't have the same equivalence meaning we have to compromise

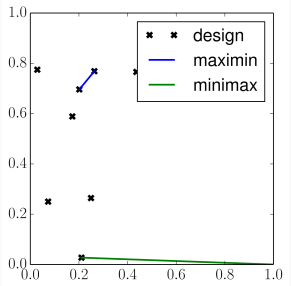
Space fillings Designs

Space filling designs

Model agnostic design

- good space coverage
- good projections to subspaces
- flexible number of evaluations

Space filling designs



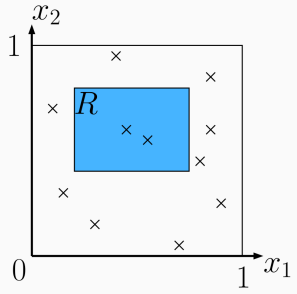
Maximin the minimum distance between two points of the design should be large

$$\max_{\mathbf{X}} \left(\min_{i \neq j} (X_i, X_j) \right)$$

Minimax the maximum distance between *any* point in space and the closest design point should be small

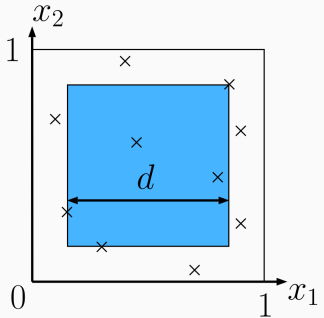
$$\min_{\mathbf{X}} \left(\max_{x \in \mathcal{D}} \left(\min_i \operatorname{dist}(x, X_i) \right) \right)$$

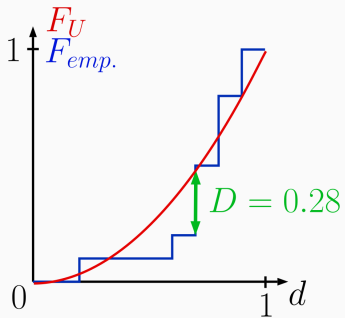
Discrepancy



Discrepancy is a measure of non-uniformity.

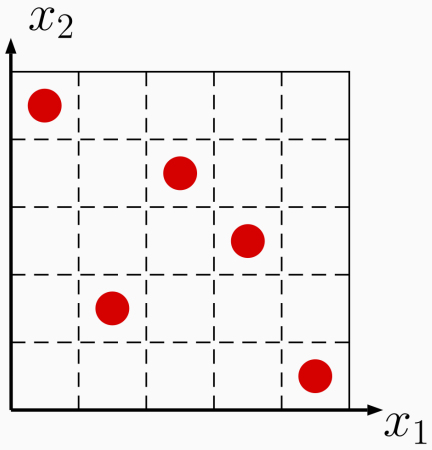
Kolmogrov-Smirnov Test





$$\text{dist}(p_{\text{emp}}(\cdot), p(\cdot)) = \sup_d(P_{emp}(\cdot), P(\cdot))$$

Latin Hypercube Design

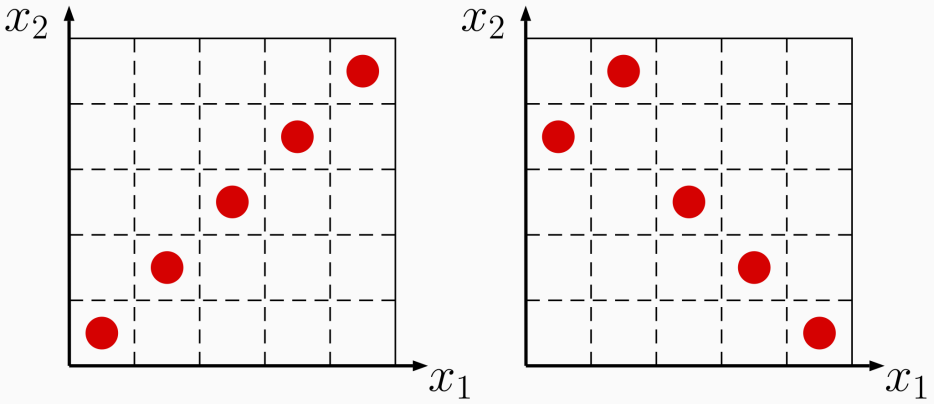


- split each dimension in n bins
- sample one point per row/column
- excellent projection properties

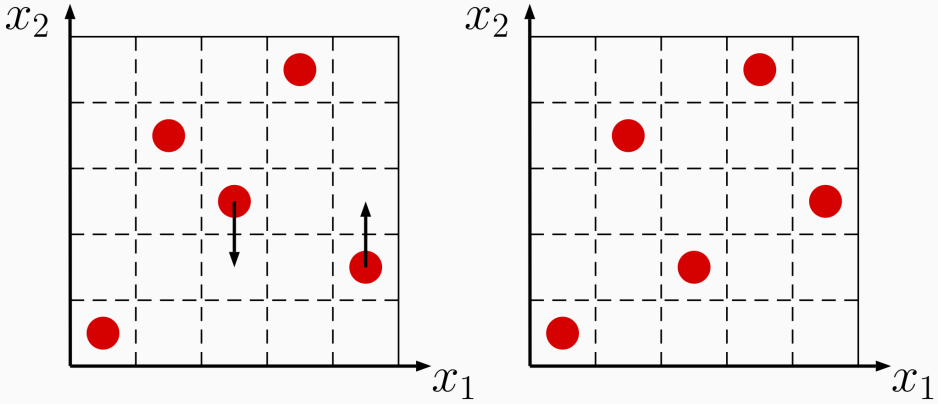
Latin Hypercube Design

4	3	1	6	7	9	5	2	8
9	6	7	2	5	8	3	4	1
5	8	2	1	4	3	9	6	7
6	5	9	8	1	7	2	3	4
3	2	8	5	6	4	1	7	9
7	1	4	9	3	2	8	5	6
8	7	3	4	2	1	6	9	5
1	4	5	3	9	6	7	8	2
2	9	6	7	8	5	4	1	3

Latin Hypercube



Latin Hypercube Optimisation



- Generate a LHD
- Evaluate `maximin` criteria for each point
- Choose a column with a ”bad” point and exchange it with a random other point
- If criterion is improved accept
- Otherwise accept with probability

$$\exp\left(\frac{\text{maximin}_{\text{new}} - \text{maximin}_{\text{old}}}{\text{iter}}\right)$$

- Construct a sequence by "reversing" the base b representation of the natural numbers
- an integer can be expressed in base b

$$a = \sum_{i=1}^{N} \beta_i b^{i-1}$$

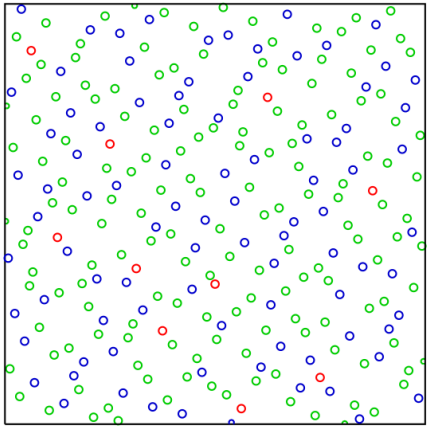
- Radical inverse function can generate a low-discrepancy sequence

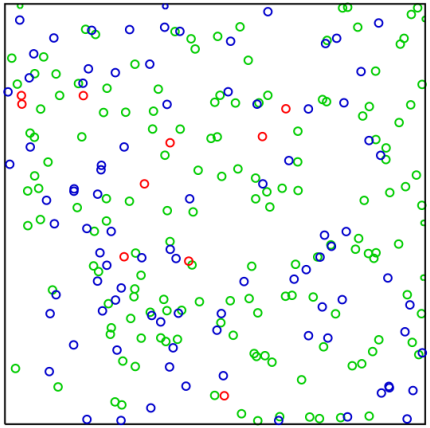
$$\Phi_b(a) = 0.\beta_1\beta_2\beta\ldots\beta_m$$

Halton Sequence (example)

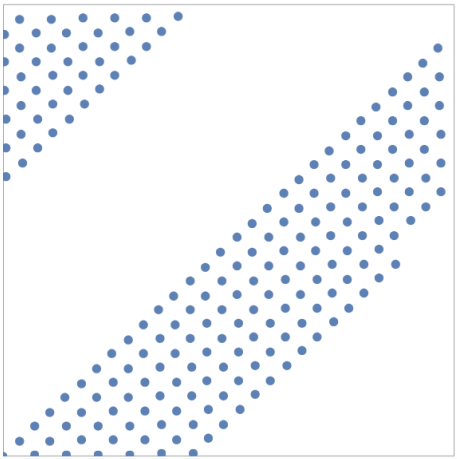
$$x_2 = \{\frac{1}{2}, \frac{1}{4}, \frac{3}{4}, \frac{1}{8}, \frac{5}{8}, \frac{3}{8}, \frac{7}{8}, \frac{1}{16}, \frac{9}{16}\}$$
$$x_3 = \{\frac{1}{3}, \frac{2}{3}, \frac{1}{9}, \frac{4}{9}, \frac{7}{9}, \frac{2}{9}, \frac{5}{9}, \frac{8}{9}, \frac{1}{27}\}$$

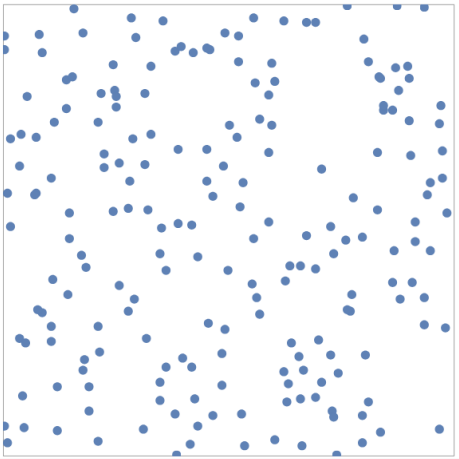
Halton Sequence





Halton Sequence [1





[1] https://www.pbr-book.org/3ed-2018/Sampling_and_Reconstruction/The_Halton_Sampler

Low Discrepancy Sequences

- Generally based on generating sequences from numbers that are co-prime
- Works well in low-dimensions
- Often regularities when numbers are large
- Include permutation or scrambling to avoid high-dimensional patterns

Summary

Summary

- Many active learning strategies are greedy, this means the initial design is very influential
- Think what you want to achieve/learn when you query data
- Design of experiments is very important and can reduce the amount of work you need to do immensely
- Biases are good and bad, be aware of them
- https://github.com/danieleongari/awesome-design-of-experiments